

Project 2

Developing an AI Application

This project not only guides you through building an AI application but also immerses you in the full product development lifecycle: from ideation to deploying an interactive AI system, and collaborating effectively within a team.

1 Introduction

In this project, students work collaboratively as a team of motivated and creative individuals to conceptualize and develop an innovative AI application. Each group is encouraged to freely select the problem they want to solve and the AI techniques they wish to apply (e.g., classification, recommendation, chatbot, computer vision, etc.). Teams should design a unique and meaningful solution that reflects their technical skills, creativity, and problem-solving mindset.

Keep in mind that that this project **does not require building AI models from scratch**. Therefore, you are encouraged to use existing frameworks, pretrained models, and APIs such as:

- Hugging Face: <https://huggingface.co/>
- Google Cloud AI: <https://cloud.google.com/ai>
- OpenAI API: <https://platform.openai.com/>
- TensorFlow Hub: <https://tfhub.dev/>
- PyTorch Hub: <https://pytorch.org/hub/>
- etc.

In addition, here are some suggested tools to support teamwork:

- Zoom: <https://zoom.us/>
- Slack: <https://slack.com/>
- Trello: <https://trello.com/>
- Discord <https://discord.com/>
- etc.

2 Required Deliverables

2.1 AI Application

Your AI application must satisfy the following requirements:

- A clear and meaningful problem definition relevant to real-world use cases.
- A system that demonstrates the capabilities of your AI application (prediction, classification, generation, detection, etc.).
- A simple and user-friendly interface (web, desktop, or mobile). You may use tools such as [Gradio](#), [Streamlit](#), [Flask](#), [Tkinter](#), [Flutter](#), etc.

2.2 Report

Prepare a detailed report that includes:

- **Project Planning and Task Distribution:** Document the team member responsibilities, which includes information on each task assigned to team members and the completion rate. E.g., Student A has percentage of completion 90% and the group work has total score of 9.0, then A receives a score of $9.0 * 90\% = 8.1$.

Therefore, it is important to evaluate the contribution of group members fairly.

- **System Design and Architecture:** Describe model selection, system components, and main functionalities. Include diagrams or figures where helpful.
- **Dataset and Models:** Briefly describe the chosen pretrained model or API.
- Include any references in a properly formatted bibliography section.

2.3 Slide

- Make clear slides that show the main points of your project.
- Prioritize images and avoid too much text.

2.4 Presentation

- All members must participate in the presentation. You may use either English or Vietnamese.
- The presentation should be coherent, impressive, and fit within a 10-minute time limit.

3 Submission

Your report and slide must be submitted in a compressed ZIP file (.zip extension) named according to the format **TeamName.zip**. If the compressed file is larger than 20MB, you may upload it to Google Drive and provide a text file with the public shared link (absolutely **no modifications** are allowed after the deadline).

Here is a detail of the directory organization:

```
TeamName
├── Report.pdf
└── Slide.pdf
```

4 Assessment

No.	Details	Score
1	AI Application	5 points
2	Report	2 points
3	Slide	1 points
4	Presentation	2 points
	Total	10 points

5 Notices

Please pay attention to the following notices:

- This is a **GROUP** assignment. Each group has 4 - 5 members.
Groups with fewer or more members than the specified limit require lab instructor approval.
- Duration: approximately 3 weeks.
- AI tools are **not restricted**; however, students should use them wisely. Lab instructors have the right to conduct additional oral interviews with random groups to assess their knowledge of the project.
- **Any form of plagiarism, dishonesty, or misconduct will result in a grade of zero for the course.**

The end.